

Tatum Abbruscato:

- Board design: designed on on-shape and made with the laser cutter.
- Game piece design: designed on on-shape and printed a set of pieces as a possible design option (final design)
- Middle transfer piece design: completed design on on-shape then printed at the EDC.
- Scoring device design: completed design on on-shape then printed at the EDC.
- Found middle arm coordinates via the joystick code to plug into Alvaro's developed code

Conner Smith:

- Claw design: designed and printed a new set of claws for the final design.
- Game pieces design: designed and printed a set of pieces as a possible design option (not final design).
- Right most arm function: worked with the left arm to find the positions where it has to go and improve the functionality of the arm.

Alvaro Cruz Perez:

- Code design: wrote the base code for the arms.
- Game pieces design: designed and printed a set of pieces as a possible design option (not final design).
- Claw design: designed a new set of claws as a possible design (not final design).
- Middle transfer piece design: Helped with the design on on-shape.
- Left arm function: worked with the left arm to find the positions where it has to go and improve the functionality of the arm.

Finally, we worked together to coordinate the arms and find the best strategy to play the game.

Project Software

Left arm code - Alvaro Cruz

```
gamecode.py * x
1 import mearmgame
2 from pynput.keyboard import Listener
3 import time
4 arm = mearmgame.meArm()
5 arm.begin(0, 0x70) # address of motor controller
6
7 x = 0 ## original x coordinate position, change as needed
8 y = 180 ## original y coordinate position, change as needed
9 z = 90 ## original z coordinate position, change as needed
10
11 arm.gotoPoint(x,y,z)
12
13 def on_press(key):
14     var = str(format(key))
15     p='\p\'
16     o='\o\'
17     i='\i\'
18     k='\k\'
19     j='\j\'
20     d='\d\'
21     q='\q\'
22     w='\w\'
23     r='\r\'
24     if var == p: #Grabs the piece at the bottom of the stacking rod
25         arm.gotoPoint(-23,180,90)# Aligns the claw with the rod and prevent the arm crashing with the rod
26         arm.gotoPoint(-23, 92, 16)
27         arm.closeGripper()
28     elif var == o: #Grabs the piece in the middle position of the stacking rod
29         arm.gotoPoint(-23,180,90)
30         arm.gotoPoint(-23, 125, 35)
31         arm.closeGripper()
32     elif var == i: #Grabs the top piece from the satcking rod
33         arm.gotoPoint(-23,180,90)
34         arm.gotoPoint(-23, 145,55)
35         arm.closeGripper()
36     elif var == k: # Stacks the grabbed piece from the transport basket into the stacking rod
37         arm.gotoPoint(100, 200,90) #The intermediat points between initial and final positions were added to avoid obstacles and smoother movement
38         arm.gotoPoint(75, 185,90) #The arm lifts the piece from the transport basket and then moves
39         arm.gotoPoint(-23, 200, 65)#The arm aligns with the rod and the stars going down
40         arm.gotoPoint(-23,170,100)
41         arm.gotoPoint(-23, 155, 70)#Fnal position
42         arm.openGripper()# Drops the piece
43     elif var == j: #This key will move the grabbed piece to the transport basket from the stacking rod
44         arm.gotoPoint(-23,155,70) # Moves the piece out of the stacking rod
45         arm.gotoPoint(-23,200,90)
46         arm.gotoPoint(40,190,90)#The next 3 points place the arm claw on top the basket without touching it
47         arm.gotoPoint(75, 185,90)
48         arm.gotoPoint(100, 170,90)
49         arm.gotoPoint(90, 165,95)
50         arm.gotoPoint(97, 145, 95)#Final position
51         time.sleep(.5)# Stops the motion of the arm
52         arm.openGripper()#Drops the piece
53     elif var == r: # Special key to pickup the sphere from the basket
54         arm.gotoPoint(-29,155,70)# Everything is the same as j but the final position is different
55         arm.gotoPoint(-20,200,90)# The last positon is different because only the sphere need to be pick up from a lower place
56         arm.gotoPoint(40,190,90)
57         arm.gotoPoint(75, 185,90)
58         arm.gotoPoint(100, 170,90)
59         arm.gotoPoint(90, 165,95)
60         arm.gotoPoint(95, 150, 87)# Final position
61         arm.closeGripper() #Graps sphere
62     elif var == d: #This key will drop the grabbed piece in the dropping zone
63         arm.gotoPoint(0,150,90)
64         arm.gotoPoint(0,200,60)
65         arm.gotoPoint(0, 160,20)
66         arm.gotoPoint(0, 85, 3)
67     elif var==q: #Opens gripper
68         arm.openGripper()
69     elif var==w: #Closes gripper
70         arm.closeGripper()
71     pass
72 def on_release(key):
73     pass
74
75 with Listener(on_press=on_press, on_release=on_release) as listener:
76     listener.join()
```

Middle arm code -Tatum Abbruscato

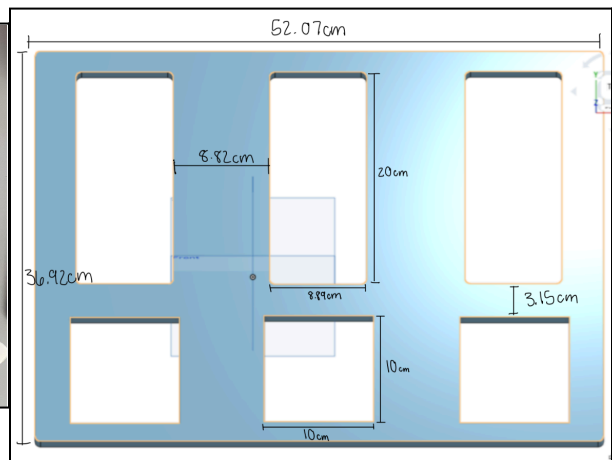
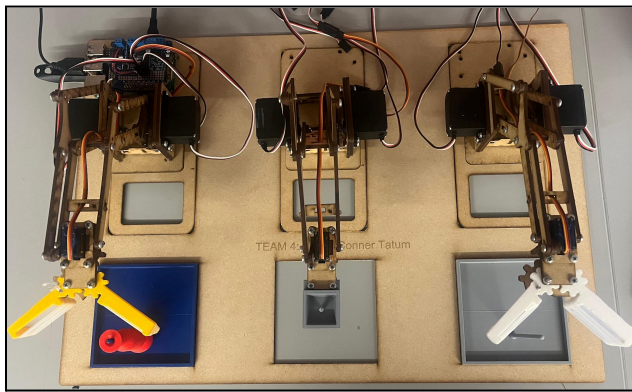
```
1 import meArm
2 from pynput.keyboard import Listener
3
4 arm = meArm.meArm()
5 arm.begin(0, 0x70) # address of motor controller
6 x = 0 ## original x coordinate position, change as needed
7 y = 60 ## original y coordinate position, change as needed
8 z = 50 ## original z coordinate position, change as needed
9
10 arm.gotoPoint(x,y,z)
11
12 def on_press(key):
13     var = str(format(key))
14     p='\p\'
15     o='\o\'
16     i='\i\'
17
18     if var == p:
19         arm.gotoPoint (-120,174,55) #sweep arm to users right
20     elif var == o:
21         arm.gotoPoint (120,169,55) #sweep arm to users left
22
23     elif var == i:
24         arm.gotoPoint(0, 50,60) #return to center
25
26     pass
27
28 def on_release(key):
29     pass
30
31 with Listener(on_press=on_press, on_release=on_release) as listener:
32     listener.join()
```

Right arm code - Conner Smith

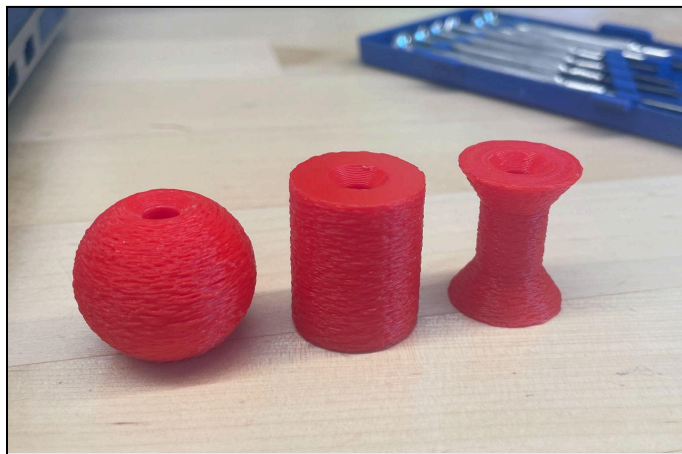
```
1 import mearmgame
2 from pynput.keyboard import Listener
3
4 arm = mearmgame.meArm()
5 arm.begin(0, 0x70) # address of motor controller
6
7 x = 0 ## original x coordinate position, change as needed
8 y = 60 ## original y coordinate position, change as needed
9 z = 50 ## original z coordinate position, change as needed
10
11 arm.gotoPoint(x,y,z)
12
13                                     #KEY BINDINGS
14 def on_press(key):
15     var = str(format(key))
16     a='a'
17     s='s'
18     d='d'
19     f='f'
20     g='g'
21     h='h'
22     d='d'
23     q='q'
24     w='w'
25
26                                     #ARM POSITIONS
27 #top piece position
28     if var == a:
29         arm.gotoPoint(57,182,80)
30         arm.gotoPoint(57,182,55)
31 #middle piece position
32     elif var == s:
33         arm.gotoPoint(57,182,80)
34         arm.gotoPoint(57,182,32)
35 #bottom piece position
36     elif var == d:
37         arm.gotoPoint(57,182,80)
38         arm.gotoPoint(57,182,10)
39
40 #raise piece to top center position
41     elif var == f:
42         arm.gotoPoint(57,182,85)
43 #middle arm piece placement
44     elif var == g:
45         arm.gotoPoint(57,182,80)
46         arm.gotoPoint(57,182,80)
47         arm.gotoPoint(-42,182,65)
48 #middle arm piece retrieval
49     elif var == h:
50         arm.gotoPoint(-42,182,65)
51         arm.gotoPoint(-42,182,80)
52         arm.gotoPoint(57,182,80)
53
54 #Open Me-Arm Gripper
55     elif var==q:
56         arm.paropenGripper(50)
57 #Close Me-Arm Gripper
58     elif var==w:
59         arm.parcloseGripper(50)
60
61     pass
62 def on_release(key):
63     pass
64 #detect input from laptop keyboard
65 with Listener(on_press=on_press, on_release=on_release) as listener:
66     listener.join()
```

Completed Project Design

Our finalized project design featured three separate Me-Arms facing the same direction with each one equidistant from the other. All three Me-Arms featured the same base, shoulder and elbow, with the claws being the only distinct element. The outer claws utilized a customized set of grippers that we believed would be able to more effectively and efficiently manipulate the game pieces. They featured two prongs with grooves built into them, and they feature rubber bands that form something of a flexible net capable of securely accommodating the various different game piece designs. The middle arm is designed specifically for the purpose of passing the game pieces between the outer two arms, and to this end it features a small peg to keep the pieces aligned and a brim to better guide the pieces onto the peg.

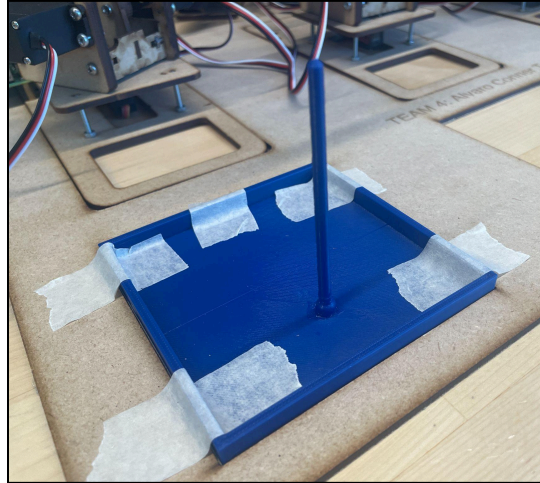


The game pieces do feature a few modifications from the standard design, while their sizes are standard they also feature slants around their openings to give our arms some clearance when manipulating them. The most noticeable feature of our game pieces is our unique texturing. When we printed our pieces we turned the fuzzy skin setting on in order to give them a more rough texture, believing this to give us more traction when manipulating them.



The final part of our design would be our game stations, of which there are few notable modifications from the standard design. In order to make a sturdier scoring rod we flared the base and made the scoring rod thicker. This was also accompanied by a short wall that would help keep the game

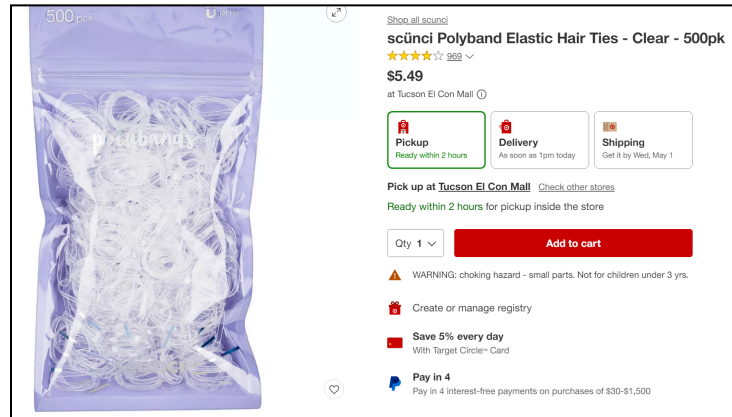
pieces inside of the scoring area if they were dropped. We also used a high infill percentage when printing them for extra strength.



As far as issues go we had many, each of which we were eventually able to overcome. The first issue that we had was more conceptual, and was of the difficulties that our regular wooden grippers would have picking up the differently shaped game pieces. Rather than setting different paths for our grippers to take for each piece we came up with our rubber band design as a “one size fits all” solution. When we put our custom grippers to practice we realized that we would have a difficult time passing our game pieces between each arm. The solution that we came up with here was our middle arm peg design, which would act as an intermediary of the two outer arms.

We also had our fair share of issues with the arms themselves. Eventually a few of our Me-Arm parts began to wear out, and the screws that we were using to hold them in place didn’t provide a tight enough fit. A solution that we came to was putting rubber bands through the loose holes to create a tighter connection. The motors that we were using also had some problems, especially the base motors. We found that the gears in the base motors would wear down quite easily and would over time be less and less accurate when following their given path. This problem had no easy solution, and the best that could be done was to disassemble our Me-Arms and replace our motors.

Bill of Materials



The only materials we purchased for our project were rubber bands for our gripper design thus a total cost of \$5.49 was spent.

Project Notes

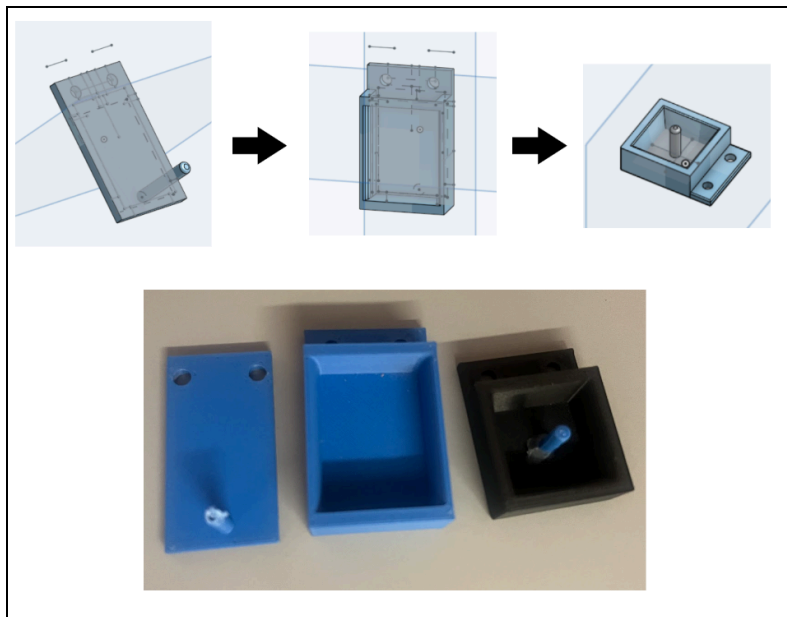


Figure 1: Transfer piece development

A notable component of our set up was the development of our middle gripper replacement. We first tried to attach a peg to place the game pieces on but found that it was difficult to align and would not be ideal for our game if it increased our chances of dropping it. We then tried a box device and found that it did not secure the game pieces as the side to side motion of the middle me-arm was not completely smooth. Lastly, we combined our two ideas to offer a secure yet easy to align device to transfer our game pieces down the line.

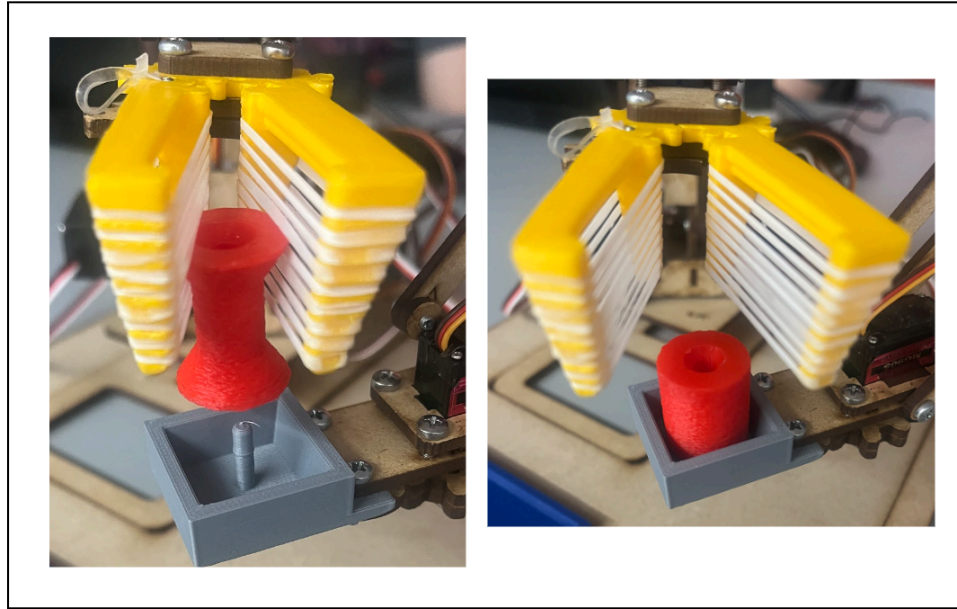


Figure 2: The handoff

The outer gripper design is notably different from the middle Me-Arm's, but is designed specifically to be able to form around and secure each of the uniquely shaped game pieces. The figure above shows how a handoff would take place using our middle box and peg combination to securely move the piece to the next arm. The wall of the box helps align the game piece on the peg, and the peg helps to hold it in place as it sweeps along the y-axis to the next arm.

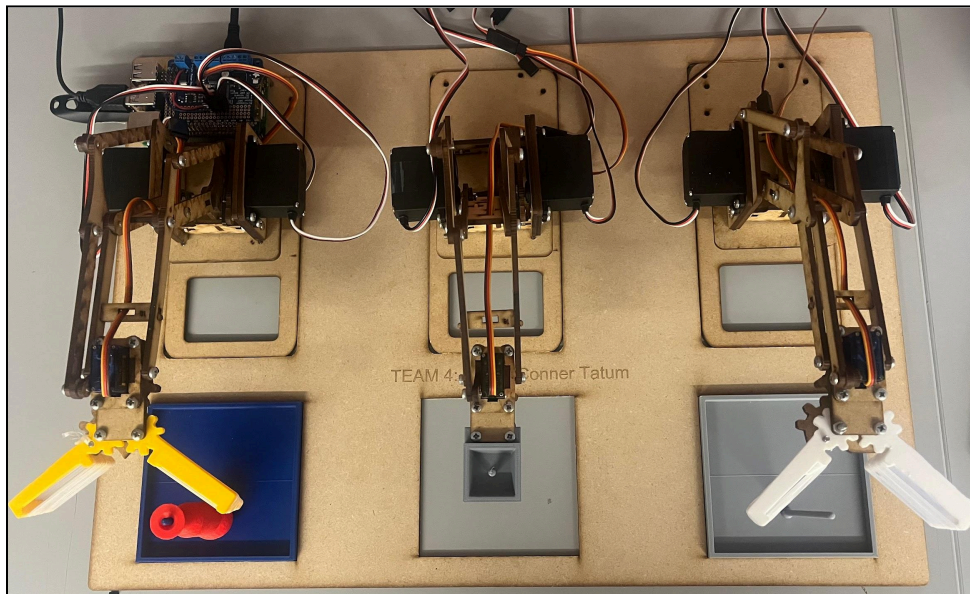


Figure 3: The final set up

We designed our board to secure the Me arms in the same direction side by side. This format worked well for passing, not having to extend the arms far for a pass because they were directly next to

each other helped save time as well as decreased the chances of dropping a piece over the distance. We made the dimensions of the board exact to the Me arm and sanded the edges of the board to create a secure fit.